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# Impedance spectroscopy as a diagnostic tool for charge transport and interface limitations in advanced silicon solar cells

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## ABSTRACT

Advanced silicon solar cell technologies employ multilayer architectures and offer significant potential to improve power conversion efficiency. However, fabrication-induced nonidealities often lead to charge-transport issues and reduced photovoltaic performance. Therefore, a systematic assessment of the device is essential to pinpoint the specific regions of performance loss and to enable targeted optimization. This study applies detailed impedance spectroscopy (IS) with a broadband AC signal in the range of 1 Hz to 1 MHz for silicon heterojunction (SHJ) solar cells; one is defect-dominated charge transfer in a-Si:H layer, another one is hindered charge transport at the p-a-Si:H/ITO hole-selective contact, and an optimized SHJ cell. These effects manifest within a distinct frequency range of the IS response, thereby enabling the identification of the dominant charge-carrier resistive and recombination-loss mechanisms. A deeper analysis of the Nyquist plot, together with frequency-dispersed phase shifts and real and imaginary impedance responses ($Z'$, $Z''$), provides clear signatures of the specific location of the performance loss. It is observed that the optimized device has a well-established depletion region and minority-carrier diffusion, with negligible resistive drop across the device. However, in unoptimized devices, additional charge-delay and impedance features appear within a specific frequency range, revealing distinct origins of the performance loss: one associated with the i-a-Si:H layer and the other with the ITO contact. Therefore, IS provides a powerful basis for diagnosing distortions in photocurrent–voltage graphs, degradation pathways, and transport bottlenecks. The frequency-resolved IS can be a critical tool for guiding interface engineering and process optimization of any optoelectronic device.





## I. INTRODUCTION

Crystalline silicon (c-Si)-based heterojunction (SHJ) solar cells that uniquely combine the proven efficiency of Si and the superior passivation and carrier-selective properties of thin hydrogenated amorphous/nanocrystalline silicon (a-Si:H/nc-Si:H) layers have emerged as a highly promising photovoltaic approach.[1,2] The core of the device features a thin intrinsic a-Si:H layer providing excellent surface passivation, which is complemented by either p- and n-type a-Si:H layers or nanocrystalline p- and n-type silicon layers that act as efficient carrier-selective contacts. A transparent conducting oxide (TCO) layer ensures lateral charge transport, also completing an optimized architecture for enhanced performance.[1–3] The unoptimized or suboptimal deposition of a single layer can lead to unwanted charge-carrier recombination and accumulation at various junctions/interfaces.[3,4] Therefore, the process and growth conditions of a-Si:H and TCO layers play a very important role in device performance.[4,5] The poor electronic quality of a-Si:H (i-, p-, n-) and TCO layers could be responsible for the band offsets and discontinuities at the interfaces with c-Si wafer, which can lead to energy barriers that hinder carrier transport across the junction, leading to enhanced bulk recombination and diminished open-circuit voltage ($V_{oc}$).[3–6] Additionally, this leads to an increase in series resistance, which can impede carrier extraction and ultimately reduce the device's fill factor (FF) and overall power conversion efficiency (PCE), resulting in S-shaped light-current density–voltage (J–V) characteristics.

To diagnose carrier-transport and recombination issues, impedance spectroscopy (IS) is capable of probing the dynamic response of the device with a small AC voltage (10–20 mV) superimposed on a steady DC bias voltage.[7] The DC bias establishes the operating condition of a solar cell, while the AC signal introduces frequency-dependent excitations that enable mapping of various physical locations and electrical phenomena throughout the device. The application of a broad frequency range (up to 1 MHz) enables probing of various slow-to-fast processes in a device with distinct time constants related to different mechanisms.[8,9] IS was primarily employed to extract key device parameters such as series and shunt resistances ($R_s$ and $R_{sh}$), carrier lifetimes, and depletion and diffusion capacitances.[10–16] Further, attempts were made to isolate the contributions of individual junctions, including the hole-selective p-a-Si:H/n-c-Si front junction and the n-a-Si:H/n-c-Si rear junction, particularly in SHJ cells using RC elements, Warburg diffusion components, and constant phase elements (CPEs) to interpret charge-transport and recombination dynamics.[8,17,18] Moreover, a device that underperforms or exhibits S-shaped light J–V characteristics is often linked to interface mismatches, contact issues, or thin-layer electronic quality issues. Hence, electronic layer-specific contributions, as reflected in their frequency-domain signatures, remain insufficiently understood in past studies. To address these issues, the devices were systematically perturbed across a wide range of frequencies (low to high; 1 Hz to 1 MHz) and forward-bias voltages (low to high; 0.0–0.68 V). Hence, it enables probing of charge-transport and accumulation processes at different depths within a device stack. For this purpose, two S-shaped devices arising from non-optimized layers, namely, the unoptimized intrinsic thin a-Si:H layer ($S_{IUO}$) and indium tin oxide (ITO), contact ($S_{TUO}$), along with one fully optimized reference device ($S_{OPT}$). The frequency-dependent discrepancies observed in the Nyquist plots of the unoptimized manifest at distinct frequency regions, which are accompanied by pronounced dispersion in phase angle (θ), real impedance ($Z'$), and imaginary impedance ($Z''$) responses. The combined evolution of these three parameters across the entire frequency spectrum reveals distinct electrical signatures associated with specific material layers and interfaces. Each interface contributes uniquely under different biasing and frequency conditions, enabling selective probing of junction- and contact-related phenomena in the SHJ device. This layer-resolved IS analysis provides direct insight into the origins of S-shaped behavior and device performance degradation. Importantly, this methodology is broadly applicable to diagnosing layer- and interface-induced losses in various multijunction solar cell architectures, including SHJ, Tunnel Oxide Passivated Contact (TOPCon), thin-film, perovskite, and Tandem solar cells, as well as other electronic and optoelectronic devices. Performance degradation can arise from any poorly optimized electronic thin layer or hetero-junction/-contact interface. Here, we show that the IS can serve as a powerful tool to localize the root cause of electrical losses and guide targeted optimization strategies in device fabrication.

## II. EXPERIMENTAL AND CHARACTERIZATION DETAILS

The SHJ was fabricated on as-cut n-type c-Si ⟨100⟩ wafers with a thickness of $180 \pm 10\,\mu$m and a resistivity of 0.8–1.2 Ω cm. The as-cut wafers were first pre-cleaned in a piranha ($H_2SO_4$/$H_2O_2$::3:1) solution to remove primary organic and metal contaminants. The surface was pre-treated with ∼2% HF in DI water (18.2 MΩ resistivity) to maintain its hydrophobicity, and saw damage was removed with ∼30 wt. % NaOH solution (10 min at 85 °C). Pyramid texturing was performed using a ∼2 wt. % KOH solution with an additive at 75–80 °C, followed by surface smoothing post-texturing using a mixture of $HNO_3$, $CH_3COOH$, and HF at 25 °C. All prepared samples were further cleaned using the standard RCA procedure to remove any remaining inorganic and metallic impurities. The cleaned and textured samples were immediately transferred to a three-chamber PECVD system (SAMCO, Japan) for the deposition of intrinsic and doped amorphous silicon layers, following an HF dip (more details can be found in Ref. 19). SHJ solar cells were fabricated with the structure: Ag/ITO/p-a-Si:H/i-a-Si:H/n-c-Si/i-a-Si:H/n-nc-Si:H/ITO/Ag. The intrinsic and p-a-Si:H layers were deposited using radio frequency (RF) power at 13.56 MHz, whereas the n-nc-Si:H layer was deposited at a higher frequency of 27.12 MHz. Subsequently, RF magnetron sputtering was used to deposit an ∼80 nm indium tin oxide (ITO) layer from an $In_2O_3$:$SnO_2$ (90:10 wt. %) target. The silver (Ag) metal contacts were finally deposited by thermal evaporation through a shadow metal mask on the front, with a thickness of ∼1 μm, and a fully covered back contact with a thickness of ∼500 nm. In the present work, during device optimization, an S-shape J–V characteristics was first observed due to the unoptimized i-a-Si:H layer referred to as sample $S_{IUO}$, in which the effective intrinsic layer thickness on the textured Si surface is relatively high (∼12 nm) compared to the optimized structure (∼7 nm). A second S-shape was then observed in the unoptimized ITO layer, referred to as $S_{TUO}$, which exhibited a high sheet resistance of >1000 Ω/□ compared to ∼40 Ω/□ in the optimized sample. After resolving both issues, the fully optimized device was obtained, designated as sample $S_{OPT}$. The schematic of an SHJ cell is shown with different experimental conditions in Fig. 1. Current density–voltage (J–V) measurements of complete SHJ devices were conducted under standard one-sun illumination using a Class AAA solar simulator (Oriel Sol3A, Newport, USA).

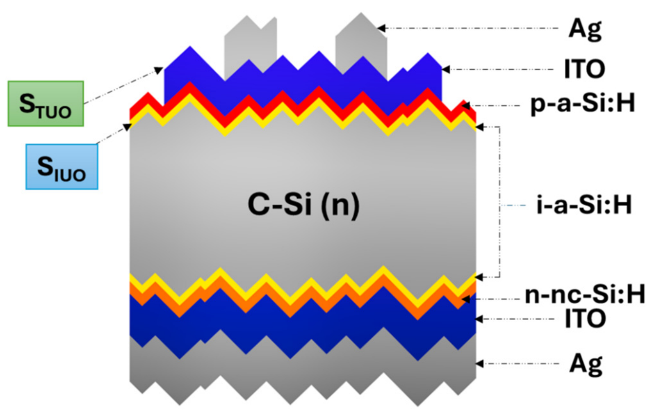


**FIG. 1.** Schematic of an SHJ solar cell structure representing experimental conditions like $S_{IUO}$, $S_{TUO}$, and $S_{OPT}$.

Series resistance-free pseudo-JV curves are recorded using Sun-$V_{oc}$ measurement on the Sinton lifetime tester (WCT 120 TS; QSSPC measurement), and the pseudo-FF and efficiency are obtained from this measurement. The biologic SP-300 system, integrated with EC-Lab software, was used for impedance spectroscopy data acquisition and analysis. CV measurement was performed and derived from impedance spectroscopy under dark conditions. A sinusoidal AC perturbation of 10–20 mV was superimposed on the DC bias, with the frequency varied from 1 Hz to 1 MHz and a precise voltage resolution of 50 $\mu$V, with the cell temperature maintained at 25 °C.

## III. RESULTS AND DISCUSSION

### A. Current density–voltage (J–V) characteristics

Figure 2 presents a comparative analysis of the illuminated J–V characteristics and corresponding Suns-$V_{oc}$ curves (pseudo-J–V characteristics, which are resistance-free) for the three SHJ devices $S_{IUO}$, $S_{TUO}$, and $S_{OPT}$. These measurements provide valuable insights into the device's resistive losses under real operating conditions.[6]

The $S_{IUO}$ device exhibits a substantial divergence between the Suns-$V_{oc}$ and illuminated J–V characteristics [Fig. 2(a)], hence indicating pronounced non-idealities under operating conditions. $S_{IUO}$ shows the lowest Suns-$V_{oc}$ (~676 mV) and a reduced illuminated $V_{oc}$ of ~671 mV, accompanied by strong curvature in the forward-bias region of the J–V curve. This behavior is reflected in its poor FF (~46.64%), high $R_s$ (~7.28 $\Omega\cdot$cm$^2$), and a markedly reduced PCE of ~10.66% (Table I), despite a relatively high pFF (~86.2%). The large disparity between pFF and FF suggests resistive losses and inefficient carrier transport. In contrast, the $S_{TUO}$ device exhibits a higher Suns-$V_{oc}$ of ~739 mV with a correspondingly high pFF of ~85.2%, indicating strong intrinsic junction quality and low recombination under open-circuit conditions [Fig. 2(b)]. However, the illuminated J–V curve shows an early voltage drop and significant FF degradation (~62.72%), arising from pronounced $R_s$ losses (~4.56 $\Omega$ cm$^2$). Consequently, the achievable PCE is limited to ~17.35%, despite the favorable Suns-$V_{oc}$ response. Among the three devices, $S_{OPT}$ shows the smallest deviation between the illuminated J–V and Suns-Voc curves [Fig. 2(c)], thereby confirming optimized fabrication with minimal resistive losses and excellent interfacial passivation. As summarized in Table I, $S_{OPT}$ delivers superior photovoltaic performance, achieving a Voc of ~735 mV, a $J_{sc}$ of ~38.65 mA cm$^{-2}$, a high FF of ~82.23%, and a PCE of ~23.35%. The close agreement between the Suns-Voc (736 mV) and illuminated Voc (735 mV) further confirms efficient charge extraction with minimal $R_s$ (0.61 $\Omega$ cm$^2$).

Overall, these observations confirm that the combined analysis of Suns-$V_{oc}$ and illuminated J–V characteristics provides a credible evaluation of solar cell output performance and enables the identification of resistive losses in the device.[6,20] However, in SHJ devices that comprise multiple functional layers and complex interfacial networks, such electrical characterization alone is insufficient to pinpoint the exact origin of performance degradation. Although both $S_{IUO}$ and $S_{TUO}$ devices exhibit clear signs of performance limitations, their illuminated J–V and Suns-$V_{oc}$ responses do not allow an unambiguous distinction between fabrication-induced losses in

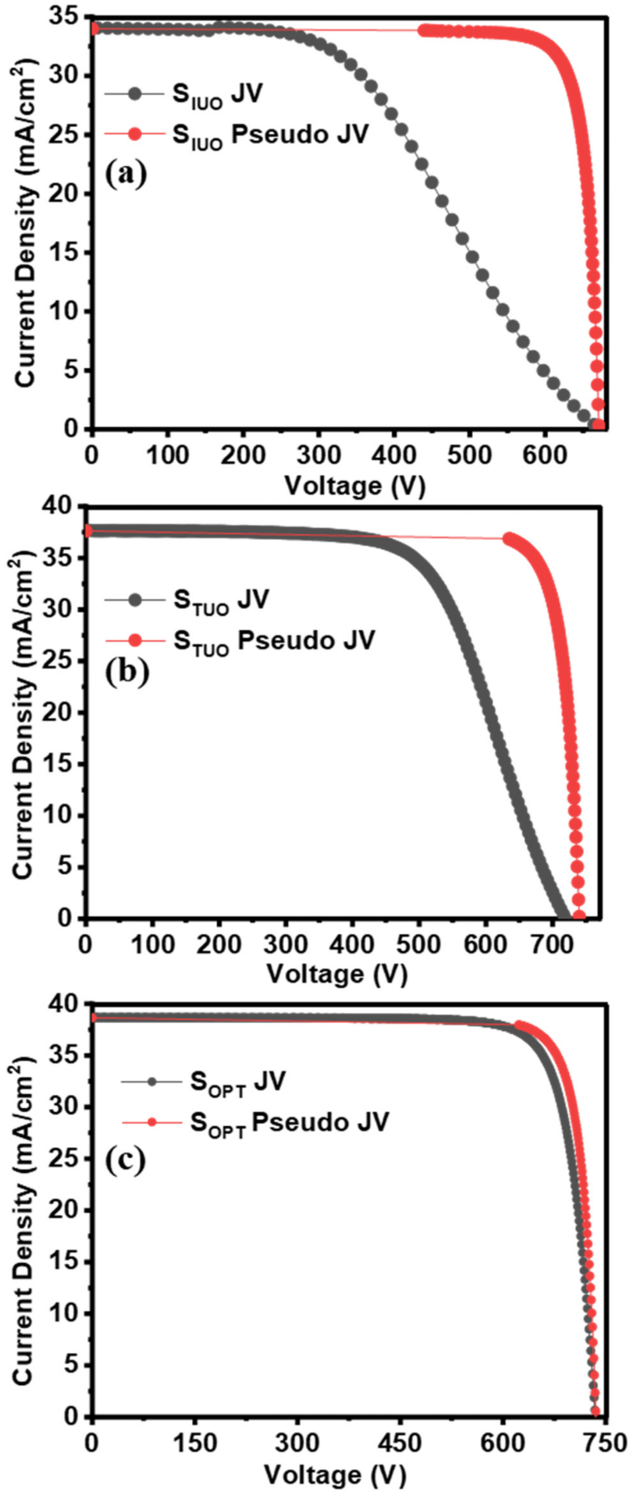


FIG. 2. Comparative analysis of illuminated J–V curves and Suns-$V_{OC}$ characteristics for the devices: (a) $S_{IUO}$, (b) $S_{TUO}$, and (c) $S_{OPT}$.

**TABLE I.** Suns-$V_{oc}$ and J–V characteristics results of devices $S_{IUO}$, $S_{TUO}$, and $S_{OPT}$.

| Samples | Suns-$V_{oc}$ (mV) | pFF (%) | $J_{sc}$ (mA/cm$^2$) | $V_{oc}$ (mV) | $R_s$ (Ω cm$^2$) | $R_{sh}$ (Ω cm$^2$) | FF (%) | PCE (%) |
|---|---|---|---|---|---|---|---|---|
| $S_{IUO}$ | 676 | 86.2 | 34.07 | 671 | 7.28 | 1200 | 46.64 | 10.66 |
| $S_{TUO}$ | 739 | 85.2 | 37.64 | 735 | 4.56 | 2300 | 62.72 | 17.35 |
| $S_{OPT}$ | 736 | 85.3 | 38.65 | 735 | 0.61 | 4500 | 82.23 | 23.35 |

the devices or the identification of the specific layer or interface responsible for the observed degradation.

### B. Nyquist plot analysis

In the present study, under the low forward bias (0.0–0.4 V), all three investigated samples ($S_{IUO}$, $S_{TUO}$, and $S_{OPT}$) exhibited a single semicircular arc in their Nyquist plots, as shown in Fig. 3. Also, the diameter of the originated semicircle is decreased with forward biasing, with a significant distortion and split in the semicircles for individual devices (Fig. 4). It is noted that at low forward bias, the impedance response is dominated by immobile charges within the depletion region. With increasing bias voltage, the depletion region narrows, and the response progressively shifts from depletion-controlled behavior to injected carrier-dominated dynamics. In which minority-carrier diffusion is the dominant process, with the injected carrier responding in large numbers. Therefore, the low-bias uniform device response arises from the dominance of the impedance associated with the depletion-region barrier, which restricts the transport of the majority charge carriers at the heterojunction interface (p-a-Si:H/i-a-Si:H/n-Si).[10,21] Here, among the three samples, $S_{IUO}$ exhibits the smallest semicircle diameter, whereas $S_{TUO}$ and $S_{OPT}$ show nearly identical diameters

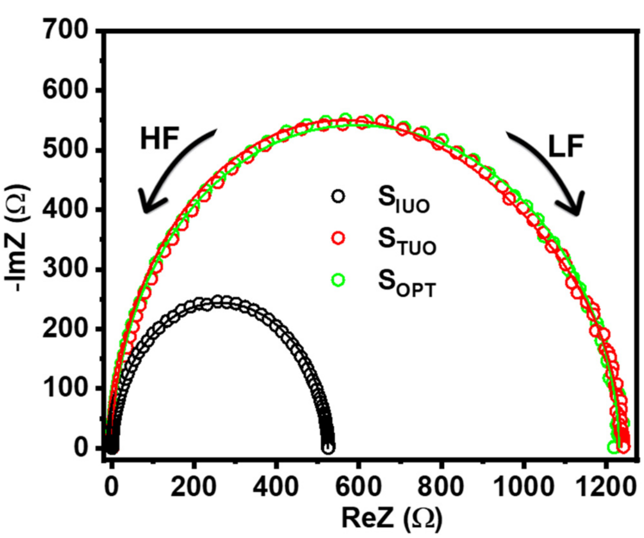


**FIG. 3.** Nyquist plots at 0.0 V bias for $S_{IUO}$, $S_{TUO}$, and $S_{OPT}$ samples, along with their respective fitting curves, HF and LF denote the high- and low-frequency regions.

(see Fig. 3). This behavior indicates that the depletion region and, consequently, the heterojunction interface are more effectively formed in $S_{TUO}$ and $S_{OPT}$ with fewer recombination states. While $S_{IUO}$ demonstrates a comparatively less effective depletion-region formation and a larger number of recombination states. Further, with increasing forward bias, all samples exhibit a noticeable semicircle distortion after ∼0.44 V, and the corresponding Nyquist semicircles become progressively narrower along the Im(Z) axis, as shown in Fig. 4. This shows that injected carriers dominated the diffusion-related dynamics.[22]

For biases >0.48 V, all samples show a clear deviation from a single semicircle in the Nyquist plot, which transforms into a distinct double semicircle pattern shown in Figs. 4(a)–4(c) for $S_{IUO}$, $S_{TUO}$, and $S_{OPT}$ cells, respectively. However, the shape and characteristics of the originated second semicircle (high-frequency region) are distinct for each of the three samples. These arcs can be regionally categorized into high-frequency semicircles (HF-SCs; >10 kHz) and low-frequency semicircles (LF-SCs; <10 kHz). Each sample exhibits a distinct frequency-dependent impedance response, as shown by the differences in the Nyquist plots in Figs. 4(d)–4(i). This indicates that injected carriers accumulate at different locations within the device depending on the presence of unoptimized or highly resistive interfaces. For $S_{IUO}$, the initial distortion in the impedance response is predominantly observed in the LF region, as shown in Fig. 4(d), whereas $S_{TUO}$ exhibits an anomalous HF semicircle behavior [Fig. 4(e)]. In contrast, $S_{OPT}$ displays a single, well-defined semicircle dominating the entire bias range up to 0.58 V [Fig. 4(f)] with a very short diameter of the HF semicircle. As the forward bias approaches 0.68 V, the distortions in the non-optimized devices become increasingly pronounced. For $S_{IUO}$, forward biasing beyond 0.58 V leads to a clear shift of the impedance response toward lower frequencies, ultimately resulting in the splitting of the LF semicircle into two distinct arcs and the emergence of a third semicircle ($LF_2$) [Fig. 4(g)]. In contrast, $S_{TUO}$ exhibits a pronounced enlargement of the HF semicircle with increasing bias [Fig. 4(h)]. At biases exceeding 0.6 V, the HF semicircle becomes dominant and effectively suppresses the LF semicircle associated with the heterojunction interface, as clearly observed at 0.68 V for $S_{TUO}$ [Fig. 4(h)].

Notably, $S_{OPT}$ shows a negligible emergence of additional semicircles even at high forward bias [Fig. 4(i)]. Therefore, the corresponding electronic equivalent circuit (EEC) modeling differs for all three samples. $S_{TUO}$ and $S_{OPT}$ are well-fitted using a double RC configuration across the entire bias range, whereas $S_{IUO}$ requires a triple RC equivalent circuit, as shown in Fig. 5. Also, here, to account for the frequency dispersion and inhomogeneous responses, the constant phase elements (CPEs) have been applied for EEC instead of ideal capacitors (see Fig. 5).[8] The impedance

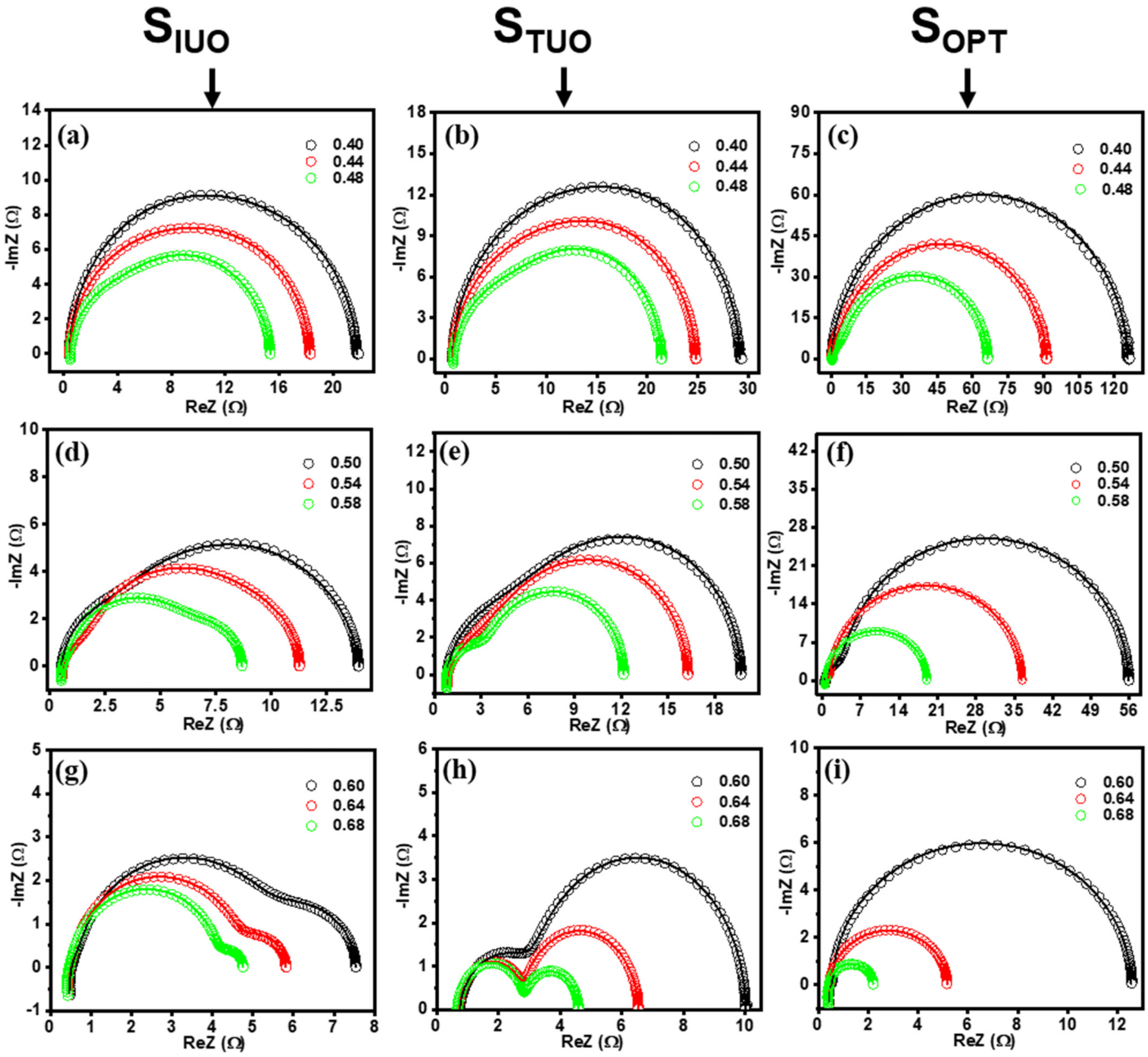


FIG. 4. Nyquist plots with fitted curves for impedance spectroscopy measurements under varying bias voltages from 0.40 to 0.48 V for three samples: (a) $S_{IUO}$, (b) $S_{TUO}$, and (c) $S_{OPT}$. Panels (d)–(f) correspond to samples $S_{IUO}$, $S_{TUO}$, and $S_{OPT}$, respectively, under extended bias conditions from 0.50 to 0.58 V. Panels (g)–(i) show the impedance response for $S_{IUO}$, $S_{TUO}$, and $S_{OPT}$ under higher bias voltages from 0.60 to 0.68 V.

Z (ω) of such a fitted circuit is given by[8]

$$Z(\omega) = R_s + \frac{R_{p1}}{1 + j\omega R_{p1} C_{p1}} + \frac{R_{p2}}{1 + j\omega R_{p2} C_{p2}} + \frac{R_{p3}}{1 + j\omega R_{p3} C_{p3}}. \quad (1)$$

Here, $R_s$ is the series resistance, while $R_P$ and $C_P$ represent the parallel resistance and capacitance components, respectively. The "ω" represents the frequency of the impedance measurement, and the subscripts 1, 2, and 3 represent the individual components representing different regions inside the device and contributing to the impedance. Here, the first terms $C_{p1}$; j $(\omega Q)^{\alpha}$ are fitted using CPE, where "α" is the dispersion coefficient and Q is related to charge storage. Accordingly, it is evident that the nonidealities in the solar cell performance due to irregularities in charge dynamics are

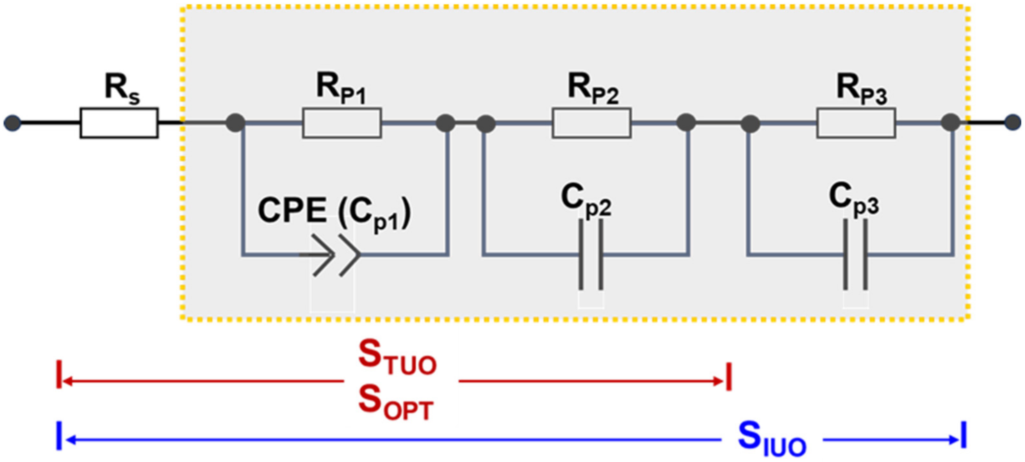


**FIG. 5.** Equivalent circuit model used for fitting the impedance data of samples: $S_{IUO}$, $S_{TUO}$, and $S_{OPT}$.

detected as depressed semicircles, distorted arcs, or the presence of multiple overlapping arcs, as shown in Fig. 4.[21,22]

These variations suggest that the origins of resistance and capacitance, and their associated time constants, differ across the samples. Therefore, the Nyquist plot can broadly identify the frequency regions where non-ideal behavior occurs. However, it presents a highly condensed, low-magnification view of the device response. Consequently, it alone cannot quantitatively separate the roles of individual interfaces and layers in the SHJ architecture, which remain unresolved. Hence, to obtain a more detailed understanding, the impedance response must be analyzed as a function of frequency. The frequency-resolved analysis of the phase shift (Bode plot), together with the $Z'$ and $Z''$ responses, could enable a more detailed comparison among devices, as examined in Secs. III C–III E.

### C. Bode plots and phase response

The Bode plots of the device represent the phase shift of the current response relative to the applied AC voltage.[23] The various layers and interfaces within the device, particularly those affected by defect states and interface band bending, influence the dynamics of charge carriers and thereby govern the phase shift of the output current.[8] Capacitive charge accumulation at these interfaces induces a phase lag in the current response, resulting in a downward (negative) phase shift in the Bode phase plot. Also, pure resistance does not cause a phase shift; however, it usually slows charge-carrier movement.[23]

Consequently, the Bode phase response provides characteristic signatures of the depletion-field-dominated regime at low bias and the accumulation of injected carriers due to diffusion and different internal interfaces under higher bias conditions.[23,24] The phase shift is defined as

$$\theta = \tan^{-1}\frac{Z''}{Z'}, \tag{2}$$

where $\theta(\omega)$ denotes the phase angle. From Eq. (2), it can be inferred that $\theta(\omega)$ indicates the relative contributions of the reactive and resistive elements in the system. Hence, each local minimum or phase dip in the phase angle corresponds to a distinct relaxation process characterized by a specific time constant $\tau_f$ related to its characteristic frequency f,[23]

$$\tau_f = \frac{1}{2\pi f}. \tag{3}$$

Therefore, the phase dip shift reflects variations in the associated time constant, indicating the quantitative delay of the corresponding transport mechanism. Also, the position on the frequency scale identifies the specific interface or layer responsible for charge accumulation in the SHJ device. Hence, each semicircle in the Nyquist plot can be assigned to a specific electrical process within the device.[23,24] Figure 6 represents the Bode phase spectra for devices $S_{IUO}$, $S_{TUO}$, and $S_{OPT}$. A prominent, well-defined phase-minimum feature at 0.0 V bias in $S_{TUO}$ and $S_{OPT}$, as shown in Fig. 6(a), indicates strong depletion formation.[25] As the forward bias increases, the phase minimum becomes significantly distorted as illustrated in Figs. 6(b)–6(d). It is attributed to the effective diffusion of minority carriers accompanied by the activation of resistive and capacitive pathways under forward biasing.[9,26,27] Notably, at 0.5 V, $S_{OPT}$ still exhibits a pronounced LF phase dip ($\theta \sim -60°$ at ~10 kHz) and HF phase dip (at >100 kHz) [see Fig. 6(b)]. Meanwhile, in $S_{IUO}$ and $S_{TUO}$, the LF minimum is considerably less dominant ($\theta \sim -40°$ at ~10 kHz) [see Fig. 6(b)]. This diminished LF response in $S_{IUO}$ and $S_{TUO}$ can be attributed to junction-driven effects where the onset of charge-transport activates secondary impedance pathways and gives an abrupt and inefficient diffusion of minority carriers.[8,23] Also, with increasing forward bias, the LF diffusion-related phase dip for $S_{OPT}$ and $S_{TUO}$ shifts progressively toward lower frequencies, from ~100 kHz at 0.5 V to ~1 kHz at 0.58 V and ~500 Hz at 0.68 V. This trend corresponds to an increasing time constant and indicates more stable minority-carrier

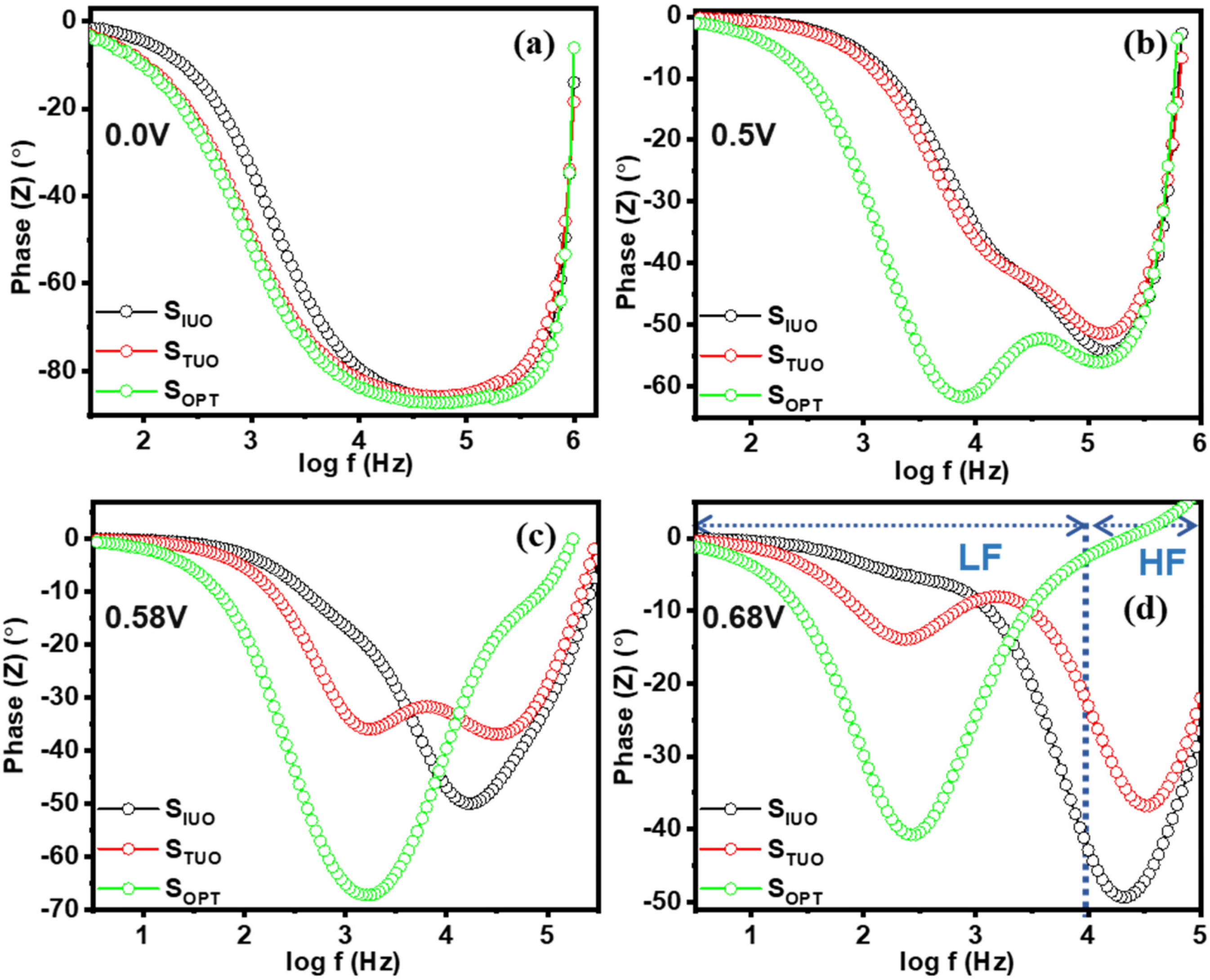


FIG. 6. Bode plots showing the frequency-dependent impedance phase angle with corresponding trends for the three samples ($S_{IUO}$, $S_{TUO}$, and $S_{OPT}$) at different applied bias voltages of (a) 0.0, (b) 0.5, (c) 0.58, and (d) 0.68 V.



accumulation in $S_{OPT}$ and $S_{TUO}$ due to low recombination pathways with increased injection.[22] In contrast, $S_{IUO}$ exhibits an almost negligible LF phase dip response. Also, at 0.58 V and beyond, $S_{IUO}$ and $S_{TUO}$ exhibit markedly different behavior compared to $S_{OPT}$. Specifically, $S_{OPT}$ maintains a well-defined LF minimum [see Fig. 6(c)]. $S_{IUO}$, in contrast, exhibits a splitting of the LF phase minimum into two components. The first appears at ∼100 Hz, consistent with the minority-carrier diffusion response ($LF_1$: minority diffusion), which is located close to the LF position observed in the other cells. The second component ($LF_2$) shows a pronounced shift toward higher frequency, emerging at ∼10 kHz, and is attributed to a secondary charge accumulation site. With increasing forward bias, this effect becomes more pronounced as the depletion region narrows and resistive interfaces increasingly dominate the injected charge response. Meanwhile, $LF_1$ evolves toward a shallow minimum or plateau, whereas $LF_2$ becomes more capacitive, showing a deeper phase dip and a significant phase shift. This behavior further indicates an additional phase delay due to charge accumulation at defective sites rather than minority-carrier diffusion.[28] Although $S_{TUO}$ also exhibits a weak response of the LF phase dip at 0.58 V due to reduced minority-carrier diffusion. Instead, the HF response (>10 kHz) becomes more dominant, with a stronger phase dip, suggesting that the injected carriers are trapped or accumulated at contact interface regions before contributing to minority-carrier diffusion.[29] Hence, the Bode phase response initially reveals the dominance of different frequency regimes where injected charges experience obstruction along their transport path within the SHJ device. Since the phase response reflects the interplay between resistive and capacitive components across different regions of the device, a clearer understanding can be obtained by analyzing the frequency-dependent $Z'$ and $Z''$ responses, which enable precise identification of the underlying charge-transport and accumulation dynamics.

### D. Real ($Z'$) and imaginary ($Z''$) parts of devices' impedance spectra analysis

The $Z'$ vs frequency response provides a critical diagnostic tool for separating the individual resistive components as a cumulative sum of series ($R_s$)+ charge transfer ($R_{ct}$)+ recombination ($R_{rec}$).[30] Here, $R_s$ represents ohmic losses arising from the electrodes and bulk materials such as the Si wafer. Moreover, a high $R_{rec}$ reflects effective passivation within the device and therefore suppresses carrier recombination at defect states, leading to enhanced minority-carrier accumulation in the quasi-neutral region.[30] In contrast, $R_{ct}$ originates at various internal interfaces due to poor material quality and consequently disturbs tunneling through the a-Si:H layers and other interfaces in SHJ devices.[30] At very high frequencies ($wRC \gg 1$, $Z' \sim R_s$; beyond 10 kHz), the capacitive elements of the solar cell are effectively short-circuited, and the device behavior is mainly determined by the $R_s$.[23] For the best-optimized device, the major contribution to the total resistance in the low forward bias and low-frequency range arises from $R_{rec}$. However, in devices with defective interfaces, the LF response is distorted by the additional contribution of $R_{ct}$. As forward bias increases, the dominant resistance in defective devices shifts from $R_{rec}$ to $R_{ct}$.[30] Hence, $R_{ct}$ directly probes the interface quality and serves as a critical metric for evaluating the device's carrier transport. Therefore, an optimized device requires high $R_{rec}$ together with low $R_{ct}$ and $R_s$. In the frequency domain, $R_{ct}$ is resolved at slightly lower frequencies (<10 kHz), and $R_{rec}$ appears in the low-frequency regime around ~1 kHz.

Figures 7(a)–7(c) show the frequency-dependent $Z'$ response of the devices. At 0.5 V, well below the built-in potential $Z'$ remains at its maximum value (~55 Ω) up to ~1 kHz for the $S_{OPT}$ device, whereas both $S_{IUO}$ and $S_{TUO}$ remain well below ~20 Ω, as shown in Fig. 7(a). This behavior indicates a more pronounced $R_{rec}$ in $S_{OPT}$ and leads to significant minority-carrier accumulation in the quasi-neutral region.[9,21] In contrast, the reduced $Z'$ values in the $S_{IUO}$ and $S_{TUO}$ devices suggest low $R_{rec}$ and, hence, enhanced carrier losses. As the applied bias increases to 0.6 V, the difference in $Z'$ among $S_{IUO}$, $S_{TUO}$, and $S_{OPT}$ gradually narrows [see Fig. 7(b)]. Specifically, $S_{OPT}$ exhibits a sharp reduction of ~12 Ω, while $S_{IUO}$ and $S_{TUO}$ show relatively slight decreases of ~10 and ~8 Ω, respectively, as shown in Fig. 7(b). This trend reflects the progressive activation of resistive transport pathways in $S_{IUO}$ and $S_{TUO}$ with increasing bias, along with additional voltage drops contributing to the overall $R_{ct}$. At a higher forward bias of 0.68 V and in the low-frequency range (1–100 Hz), $S_{IUO}$ and $S_{TUO}$ exhibit higher $Z'$ values of 4–5 Ω compared to $S_{OPT}$ (~2 Ω), as shown in Fig. 7(c). This behavior indicates that $R_{ct}$ becomes the dominant contributor to the voltage drop under these conditions. Furthermore, $S_{IUO}$ shows the highest $Z'$ response in the <10 kHz region (longer time scale), which suggests that the voltage drop occurs deeper within the device. Whereas $S_{TUO}$ exhibits elevated $Z'$ primarily in the >10 kHz region (shorter time scale) further indicating a combined influence of $R_{ct}$ and $R_s$.

These observations imply that the non-optimized i-a-Si:H layer in $S_{IUO}$ introduces higher resistance at low frequencies associated with deeper device regions (a-Si:H layers and wafer).[31,32] However, the enhanced high-frequency $Z'$ response in $S_{TUO}$ points to resistive losses originating from fast-response interfaces such as metal or TCO-related contacts.[32] To further resolve the individual relaxation processes and charge-carrier dynamics, the frequency dependence of $Z''$ is examined. The individual $Z''$ vs frequency response clarifies the transition between distinct capacitive regimes as a function of applied bias.[15] As shown in Figs. 7(d)–7(f), the $Z''$ response is confined to the intermediate frequency range (100 Hz to 100 kHz), where the characteristic relaxation due to carrier diffusion and interfacial charge accumulation coincides. At low forward bias, the $Z''$ spectra are dominated by depletion capacitance ($C_{dep}$) in the vicinity of 1 kHz. At high forward bias, as the barrier potential approaches saturation, the device response increasingly dominates the diffusion capacitance ($C_{diff}$), resulting in a significantly lower frequency shift (~100 Hz). It defines long-lived minority carriers in high-quality devices.[3,15] The magnitude and frequency of $Z''$ at this point serve as a fingerprint for the passivation quality of the intrinsic a-Si:H layer. Therefore, the peak at this point quantifies the minority-carrier accumulation in the device's quasi-neutral regions. With increasing forward bias, a mid-frequency capacitive feature in the 10–100 kHz range emerges in defective devices. This corresponds to the interface-state capacitance ($C_{it}$), whose characteristic frequency depends on the specific defective interface in the device.[15] It could be a true signature of any obstructed relaxation or carriers trapped in the disordered buffer layer and non-ohmic metallic interfaces, which introduces a secondary time constant. In Fig. 7(d), at 0.5 V, the $S_{OPT}$ device exhibits a pronounced peak around 1 kHz, indicating effective minority-carrier accumulation associated with $C_{diff}$ in the best-performing cell. However, $S_{TUO}$ and $S_{IUO}$ exhibit smaller peaks, indicating fewer charges accumulated in the quasi-neutral region and increased carrier loss at defective interfaces. With forward biasing at 0.58 V, all samples exhibit a weak $Z''$ peak around ~100 kHz, which indicates the emergence of a $C_{it}$. This frequency range corresponds to charge accumulation at non-ohmic interfaces with slight Schottky-like behavior.[23] Upon further biasing to 0.68 V, the optimized $S_{OPT}$ device becomes dominated by a single diffusion-related peak where the other impedance sites effectively act as conductive channels under high injection, as shown in Fig. 7(e). In contrast, both $S_{IUO}$ and $S_{TUO}$ show a stronger $C_{it}$ contribution in the impedance response, which further reflects increased loss sites.

In $S_{IUO}$, the $Z''$ peak near ~100 Hz splits into two components: P1 and P2. The P1 position corresponds to the minority-carrier diffusion response. The P2 component shifts markedly toward higher frequencies (~100 kHz) and becomes more pronounced [see Figs. 7(e) and 7(f)], which indicates that injected carriers are trapped at relatively shallow internal interfaces rather than diffusing into the quasi-neutral region. This behavior signifies

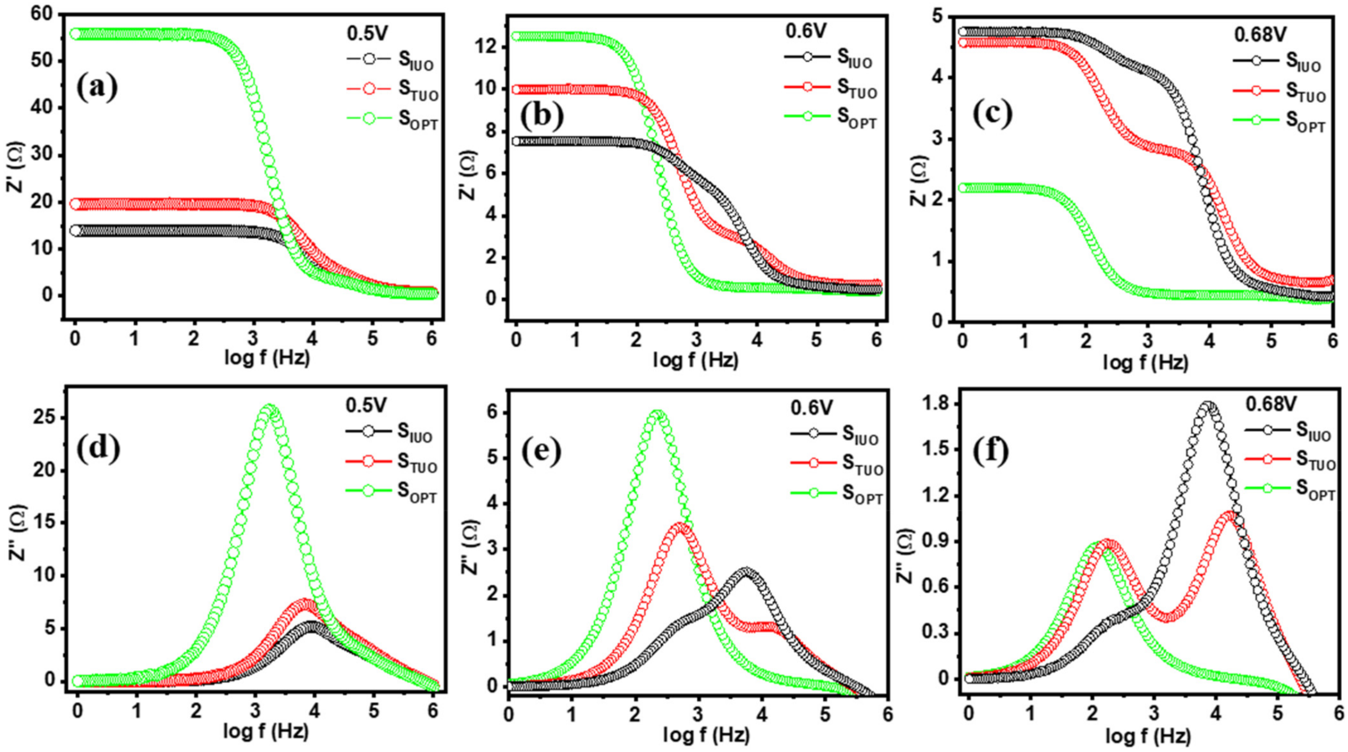


**FIG. 7.** Variation of [(a)–(c)] real part (Z′) and [(d)–(f)] imaginary part (Z″) of impedance with frequency for $S_{IUO}$, $S_{TUO}$, and $S_{OPT}$ samples at applied biases of 0.5, 0.6, and 0.68 V, respectively.



suppressed minority diffusion and enhanced accumulation of injected carriers. Moreover, in $S_{TUO}$, the high-frequency Z″ peak (>100 kHz) becomes increasingly dominant with bias, further indicating the carrier losses at fast-response interfaces, most likely associated with the ITO/p-a-Si:H contacts.[32] Thus, in both $S_{IUO}$ and $S_{TUO}$, injected carriers do not effectively diffuse as minority carriers. However, the dominant loss locations differ for $S_{IUO}$, which occurs at deeper interfacial regions and at fast metal-related interfaces in $S_{TUO}$, as evidenced by the distinct Z″ peak positions.

## E. Capacitance–voltage characteristics

The capacitance–voltage (C–V) measurements provide quantitative analysis of the device's capacitances $C_{dep}$, $C_{diff}$, and $C_{it}$.[22,27,33] Here, the $C_{dep}$ is directly related to the heterojunction interface and originated from the space charge region, represented in Fig. 8(a). It is represented by the relationship

$$C_{\mathrm{dep}} = \frac{A\varepsilon}{W} = A\sqrt{\frac{q\varepsilon_o\varepsilon_r N_d}{2(V_{\mathrm{bi}} - V)}}. \quad (4)$$

Here, $A$ is the device area, $W$ is the junction width, $V_{bi}$ is the built-in potential, $V$ is the voltage applied, $q$ is the charge of the electron, $\varepsilon_o$ and $\varepsilon_r$ are the permittivity of free space and medium, and $N_d$ is the donor carrier density. As the DC bias transitions from reverse to forward bias, the depletion width W decreases with increasing $C_{dep}$. At high forward bias, carrier injection is higher, and the device's overall charge-storage response combines mobile carrier storage ($C_{diff}$) and stored charge in traps or accumulation states ($C_{it}$).[34] Therefore, the total capacitance is given by

$$C_{\mathrm{tot}} = q^2\frac{dn_{\mathrm{stored}}}{dE_F} = C_{\mathrm{diff}} + C_{\mathrm{it}}. \quad (5)$$

Here, $n_{stored}$ is the total amount of charge stored. $E_F$ is the Fermi energy in that case, and $C_{diff}$ originates from the spatial distribution and storage of injected minority carriers within the quasi-neutral regions.[27,30] It can be expressed as

$$C_{\mathrm{diff}} = \frac{dQ}{dV} = \frac{\tau e}{2kT}I_0\exp\left(\frac{eV}{nkT}\right). \quad (6)$$

Here, $I_0$ is the diode saturation current at the applied voltage V, $\tau$ is the minority-carrier lifetime, and $n$ is the diode ideality factor. The $C_{it}$ arises from interface, trap states, and parasitic elements, including interfacial resistance and capacitance at the electrode or junction, can be expressed as[30]

$$C_{it} = q^2 g_{\mathrm{eff}}(E_F). \quad (7)$$

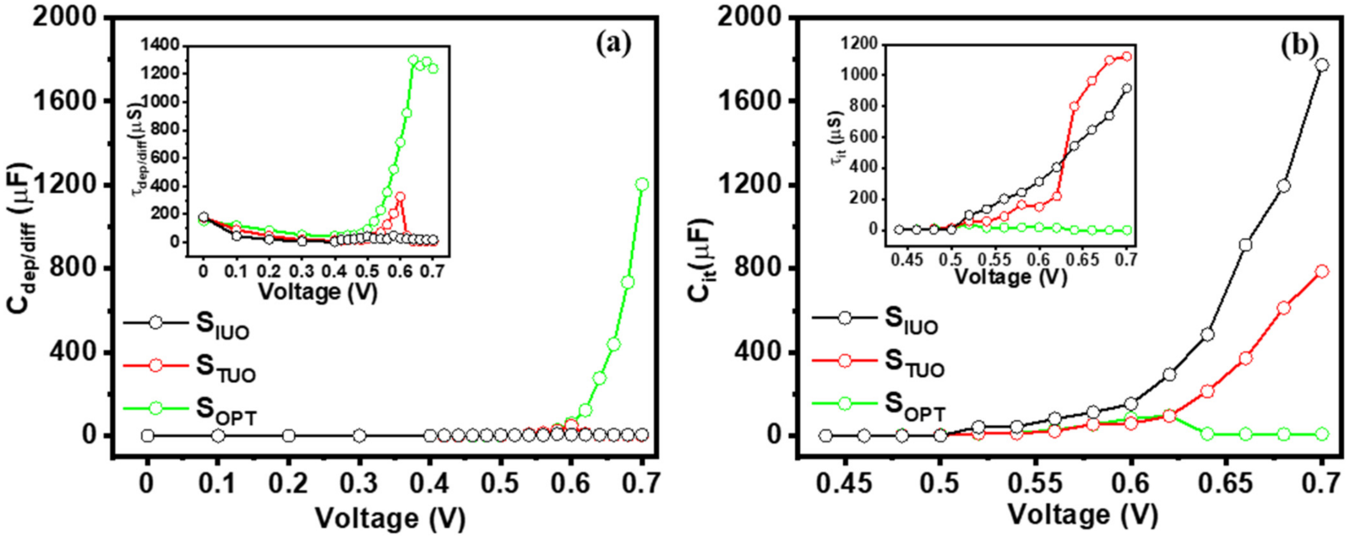


**FIG. 8.** Bias-dependent capacitance parameters extracted from impedance spectroscopy for $S_{IUO}$, $S_{TUO}$, and $S_{OPT}$ devices: (a) $C_{dep/diff}$ with inset $\tau_{dep/diff}$ and (b) interface capacitance $C_{it}$ with inset $\tau_{it}$.

The $g_{eff}$ ($E_F$) denotes the effective density of electronic states at the quasi-Fermi level, encompassing localized trap states, interface states, and injection-induced states that participate in bias-dependent charge storage.[30] If $C_{diff}$ dominates the device response, it indicates that the injected carriers possess relatively long lifetimes and exhibit simple transport and recombination dynamics, with minimal involvement of localized or interfacial charge storage. In contrast, dominance of the $C_{it}$ signifies that the injected carriers are strongly influenced by localized trap states, interface states, and injection-induced states, leading to pronounced interfacial charge storage and delayed carrier response.[8,22,32] In the present case, the dominant semicircular feature corresponds to the $C_{dep/diff}$, whereas the additional semicircular response is associated with the $C_{it}$.

Under the varying DC bias conditions, the derived RC time constants ($\tau_{dep/diff}$ and $\tau_{it}$) [see inset Figs. 8(a) and 8(b)] from both $C_{dep/diff}$ and $C_{it}$ give critical insights into the spatial localization of stored charges and the timescales of their release, etc.[26] In the $S_{IUO}$, a substantial fraction of the applied voltage drops across the interfacial i-a-Si:H layer rather than diffusion at neutral regions, giving less $C_{diff}$ of 4.21 μF and $\tau_{diff}$ of ∼25 μs [see Fig. 8(a)]. While $C_{it}$ increases steeply with forward bias, reaching 1197.07 μF at 0.68 V, it reflects significant charge accumulation and voltage drop inside the device [see Fig. 8(b)]. Consequently, charge trapping and delayed release ($\tau_{it}$ > 800 μs) prevent the diffusion of minority carriers [see the inset of Fig. 8(b)]. For the $S_{TUO}$ device, the $C_{diff}$ remains low (5.64 μF) with a $\tau_{diff}$ of (∼9 μs), while $C_{it}$ increases to 612.2 μF with a much longer $\tau_{it}$ ∼ 1200 μs [see Figs. 8(a) and 8(b)]. This behavior indicates that the highly resistive ITO contact introduces a significant charge drop at the ITO/p-a-Si:H interface rather than diffusion. In contrast, the optimized $S_{OPT}$ device shows the highest $C_{diff}$ (∼1200 μF) with minimal $C_{it}$, hence indicating that the majority of the applied injected carriers diffused as a minority rather than at parasitic interfaces. This confirms smooth charge transfer across all interfaces and efficient modulation of the depletion width under forward bias. It exhibits a much shorter $\tau_{it}$ (∼0.67 μs), reflecting faster carrier transport and minimal bulk trapping, consistent with its superior electrical performance.[9,35]



## IV. IMPEDANCE-DRIVEN PERFORMANCE DIAGNOSIS

In the present work, the distinct impedance responses of the devices enable identification of the underlying issues and provide an effective diagnosis of performance-limiting loss mechanisms. The optimized $S_{OPT}$ device exhibits two dominant impedance contributions associated with the depletion region and minority-carrier diffusion, which are essential and desirable charge-transport mechanisms in SHJ solar cells. In contrast, $S_{IUO}$ and $S_{TUO}$ show additional charge-drop sites at different locations within the device. The LF-dominated (<100 kHz) additional resistive and capacitive response observed in $S_{IUO}$ indicates that a significant fraction of injected carriers is lost at unoptimized i-layer interfaces rather than diffusing as minority carriers. In SHJ devices, charge carriers do not traverse the a-Si:H layer in a single tunneling step. Instead, the transport proceeds via trap-assisted tunneling (TAT), often described as multistep tunneling through localized tail states or midgap states. Any unoptimized/thick i-a-Si:H layer disrupts the TAT process and causes carriers to become trapped in deep defect states instead of being transported across the junction.[32] This results in additional charge-relaxation sites and an increased resistive contribution in the impedance response, leading to higher $R_s$, reduced $V_{oc}$, and enhanced carrier losses associated with lower $R_{sh}$. In $S_{TUO}$, a dominant resistive-capacitive response is observed in the high-frequency region (>100 kHz), whereas in $S_{IUO}$ it is absent. This indicates that the primary limitation arises from relatively

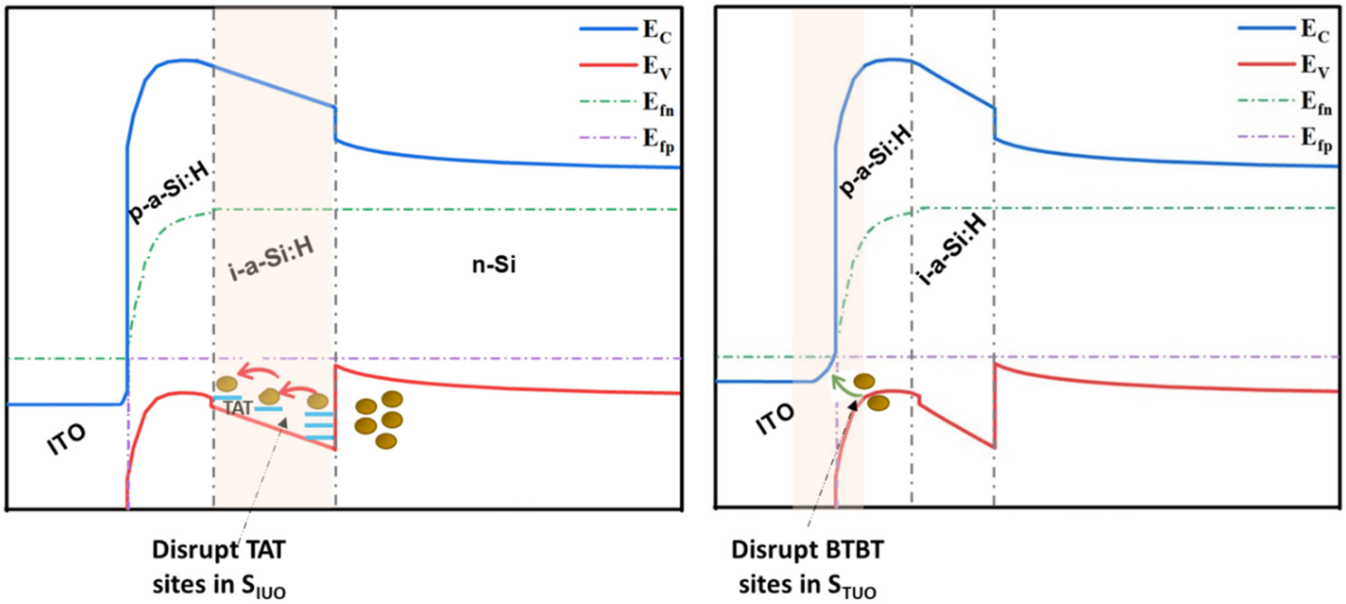


**FIG. 9.** Schematics of unoptimized SHJ devices ($S_{IUO}$ and $S_{TUO}$) energy band diagrams illustrating representative defect locations at the n-Si/i-a-Si:H interface and band alignment at the n-Si/i-a-Si:H/p-a-Si:H/TCO junction.

shallow interfaces within the device rather than deeper interfacial regions, as in $S_{IUO}$, most likely associated with a non-optimized ITO layer. The insufficient doping and mobility in the non-optimized ITO result in high sheet resistance and increased contact resistivity at the ITO(n+)/p-a-Si:H interface, thereby obstructing band-to-band tunneling (BTBT).[32] Hence, this suppresses the tunneling probability, enhances interfacial carrier losses, and limits charge extraction, resulting in a higher $R_s$. The related band structure, including the representation of disruption sites in $S_{IUO}$ and $S_{TUO}$, is shown in Fig. 9 for better understanding. These results signify that different output device parameters manifest within specific frequency windows of the impedance response, thereby explaining the macroscopic I–V characteristics. At 0 V bias, the impedance response reflects proper junction formation and depletion behavior, consistent with the device's implied Voc. With increasing forward bias, additional impedance sites become active, located in a specific frequency range. In SHJ devices, $Z'$ dominance at >100 kHz is typically associated with metal- or contact-related losses, whereas low-frequency $Z'$ (<100 kHz) reflects charge-transfer resistance arising from internal interfaces, which primarily impacts the FF of the device. Similarly, the $Z''$ provides insight into the total capacitance of the device and reflects the sensitivity of the charge storage. The behavior of $Z''$ within a specific frequency range separates the capacitive response into $C_{dep/diff}$ around ∼1 kHz and $C_{it}$ within 100 kHz. Therefore, it identifies unoptimized charge-storage sites that significantly affect quasi-Fermi level splitting in the device and thereby impact the overall $V_{oc}$. These contributions, which arise from recombination pathways, interface traps, unoptimized i-layers, or transport barriers, are distinguished by their characteristic frequency-domain signatures. Hence, the bias-induced shifting and splitting of the $Z''$ describes the emergence of multiple charge-storage sites, which effectively reduce minority-carrier diffusion and manifest as capacitive losses associated with different interfacial regions. Also, the frequency shift of diffusion-related features further reflects recombination activity and passivation quality, where stable LF responses correspond to longer carrier lifetimes and improved $V_{oc}$.



## V. CONCLUSIONS

In the present work, the origin of S-shaped J–V characteristics in two differently defective SHJ devices is diagnosed using impedance spectroscopy. A detailed analysis of Nyquist plots, combined with frequency-dispersed phase-shift, $Z'$, and $Z''$ responses, reveals that the optimized device ($S_{OPT}$) exhibits impedance contributions primarily governed by depletion and minority-carrier diffusion dynamics, which are essential for efficient device operation. From the Bode response, it is observed that the device with a non-optimized i-layer ($S_{IUO}$) is dominated by charge-transfer resistance ($R_{ct}$) in the <10 kHz frequency range. In contrast, the device with a non-optimized ITO layer ($S_{TUO}$) exhibits dominant $R_{ct}$ contributions at frequencies >10 kHz. This distinction indicates that $S_{IUO}$ experiences resistance from deeper regions within the device, whereas $S_{TUO}$ is limited by fast interfacial resistive losses at the p-a-Si:H/ITO contact. The $Z''$ response further reveals dominant capacitive charge-accumulation sites in the defective devices. In $S_{IUO}$, the diffusion-related capacitive response splits under forward bias into a reduced minority diffusion component and a high-frequency shifted peak (∼100 kHz), indicating carrier trapping before reaching the junction. In $S_{TUO}$, the high-frequency

(>100 kHz) capacitive response becomes increasingly dominant with bias, signifying carrier loss at the fast p-a-Si:H/ITO interface. Overall, this study demonstrates that each unoptimized layer or interface introduces a distinct impedance signature within a specific frequency window. By combining frequency- and bias-dependent IS, the charge dynamics can be selectively probed at different depths of the device. It provides a powerful, layer-resolved diagnostic framework for identifying the origins of S-shaped J–V behavior and performance degradation. This approach is broadly applicable to advanced multilayer silicon solar cell architectures, including SHJ and TOPCon solar cells, as well as to any heterojunction-based optoelectronic device.

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## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose.

### Author Contributions

**Ruchi K. Sharma:** Conceptualization (lead); Data curation (equal); Investigation (lead); Methodology (equal); Software (lead); Visualization (equal); Writing – original draft (lead). **Shahnawaz Alam:** Conceptualization (equal); Data curation (equal); Investigation (equal); Methodology (equal). **Silajit Manna:** Conceptualization (equal); Data curation (equal); Methodology (equal). **Son Pal Singh:** Formal analysis (equal); Visualization (equal); Writing – review & editing (equal). **Vamsi Krishna Komarala:** Formal analysis (equal); Funding acquisition (lead); Project administration (lead); Resources (equal); Supervision (lead); Validation (equal); Visualization (equal); Writing – review & editing (equal).

## DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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